

ODD POLYNOMIAL COUNTEREXAMPLES FOR THE A_d ROOT SYSTEM IN EVERY DIMENSION $d \geq 4$

ABSTRACT. For each $d \geq 4$, we give an existence proof of an odd real polynomial F of degree at most $3d(d+1) - 3$ whose restriction to the unit sphere agrees with no linear functional on any orthogonal image of the normalized A_d root system. The construction adds an arbitrarily small circle-invariant polynomial perturbation to a cubic–quintic seed from the previous dimension-four and dimension-five construction. A rank-one spectral argument bounds the dimension of every signed circle-orbit stratum of root configurations. Invariant polynomial interpolation and an incidence-dimension argument then exclude all configurations having no root in either of the two nontrivial weight planes. The remaining configurations are treated by a necessary normal form and an explicit, nonzero quintic obstruction. The proof does not require a computer-algebra certificate or a classification of all cubic solutions. We also derive the corresponding counterexamples to the regular-simplex universal-cover conjecture. The auxiliary perturbation is existential; we do not prove that the unmodified quintic seed works in all dimensions.

Verification status. This is an AI-generated manuscript prepared from a research conversation with ChatGPT. It presents a proposed proof for independent human verification; it has not been independently verified or peer reviewed. The algebraic audit supplied with the source checks selected identities, not the validity of the entire proof.

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1. STATEMENT, PROVENANCE, AND PROOF STRATEGY

Set $N = d + 1$, and give

$$V = \mathbf{1}^\perp \subset \mathbb{R}^N, \quad \mathbf{1} = (1, \dots, 1)^T,$$

its induced Euclidean metric. The normalized positive roots are

$$\alpha_{ij} = \frac{e_i - e_j}{\sqrt{2}}, \quad 0 \leq i < j \leq d. \quad (1.1)$$

Their number is $n = \binom{N}{2} = d(d+1)/2$. Normalization matters here: *all function evaluations in this paper are at unit vectors*. Let $\mathcal{O}_d$ be the space of orthogonal isomorphisms $V \rightarrow \mathbb{R}^d$. After a fixed orthogonal identification of these spaces, $\mathcal{O}_d$ is $O(d)$.

Theorem 1.1. *For every integer $d \geq 4$, there exists an odd real polynomial $F : \mathbb{R}^d \rightarrow \mathbb{R}$ of degree at most*

$$D = 6n - 3 = 3d(d+1) - 3 \quad (1.2)$$

such that no $A \in \mathcal{O}_d$ and $b \in \mathbb{R}^d$ satisfy

$$F(A\alpha_{ij}) = \langle b, A\alpha_{ij} \rangle \quad (0 \leq i < j \leq d). \quad (1.3)$$

In particular, $F|_{S^{d-1}}$ is a smooth odd counterexample even when only orientation-preserving placements are allowed.

There is a more precise perturbative conclusion. Fix the orthogonal splitting

$$\mathbb{R}^d = E_1 \oplus E_2 \oplus E_0 = \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{R}^{d-4} \quad (1.4)$$

and define

$$h_\theta(z_1, z_2, t) = (e^{i\theta} z_1, e^{2i\theta} z_2, t), \quad (1.5)$$

$$Q(z_1, z_2, t) = \operatorname{Re}(z_1^2 \bar{z}_2), \quad H(z_1, z_2, t) = |z_1|^2 \operatorname{Im}(z_1^2 \bar{z}_2). \quad (1.6)$$

Let $\mathcal{P}_d$ denote the real vector space of odd polynomials of degree at most D that are invariant under every h_θ .

Theorem 1.2. *For each $d \geq 4$, there is $\varepsilon_0 > 0$ with the following property. For every $0 < |\varepsilon| < \varepsilon_0$ and every $\eta > 0$, there exists $R \in \mathcal{P}_d$, with $\|R\| < \eta$ in any fixed norm on $\mathcal{P}_d$, such that*

$$F = Q + \varepsilon H + R$$

has no agreement of the form (1.3).

For the geometric interpretation, define

$$C_d = \{y \in V : |\langle y, \alpha_{ij} \rangle| \leq \frac{1}{2} \text{ for every } i < j\}. \quad (1.7)$$

This is the appropriately scaled polar of the difference body of a regular simplex: if $\Delta = \operatorname{conv}\{e_i - N^{-1}\mathbf{1}\}$, then $\Delta - \Delta = \operatorname{conv}\{e_i - e_j : i \neq j\}$. The distance of every facet hyperplane in (1.7) from the origin is $1/2$.

Corollary 1.3. *For every $d \geq 4$, there is a compact convex body $K \subset \mathbb{R}^d$ of constant width 1 that is not contained in $AC_d + b$ for any $A \in \mathcal{O}_d$ and $b \in \mathbb{R}^d$. Thus C_d is not a universal cover for sets of diameter 1.*

1.1. Prior work and attribution. Kuperberg [K99, Section I, Proposition 1] identifies agreement of odd functions with linear functions on rotated facet normals as the functional form of the constant-width circumscription problem. His paper proves the positive result in dimension three and explains why the corresponding homological obstruction vanishes in higher dimensions. Its Section V also uses determination of quadratic forms by root evaluations in the low-dimensional affine problem. We give the required uniform linear-algebra statement below.

The specific polynomials Q, H in (1.6), the weight- $(1, 2)$ circle action, the exact-label reduction, the cubic annihilator, and the compactness perturbation strategy are inherited from the AI-generated repository manuscript [CG26], which reports counterexamples in dimensions four and five. The VibeMathed entry [VM26] is the route by which that manuscript was supplied. We reproduce the identities needed here and do not use its dimension-specific algebraic certificates or assume the correctness of those certificates.

Chen [Ch26, Section 2] gives counterexamples for other facet configurations. More particularly, his Appendix A, Proposition 13, reproduces Makeev's finite-dimensional interpolation and dimension-count argument above the threshold $n = d(d+1)/2$. The generic-avoidance argument in this paper follows that general pattern. At the threshold, unrestricted dimension counting alone gives no strict inequality. Here we impose a circle symmetry, separate signed orbit types, and exploit the one-dimensional symmetry in every incidence fiber. The spectral fiber bound and the uniform treatment of exceptional placements are the additional arguments proposed in this manuscript.

Our only non-elementary proof input is standard semialgebraic triviality and dimension theory, recorded with references in section 3. The complete-graph cut/cycle identity is also related to the combinatorial Hodge decomposition in [JLYY11]; the elementary part used here is proved directly. No assertion of novelty is made for these background tools.

1.2. Organization. The proof has two independent parts. In section 4, generic polynomials in $\mathcal{P}_d$ exclude all placements for which no root lies in $E_1 \cup E_2$. In sections 5 and 6, the seed $Q + \varepsilon H$ excludes the complementary compact set for sufficiently small nonzero ε . Openness on that compact set permits the two parts to be combined in section 7. Section 8 proves the geometric corollary, including the support-function construction.

2. SIMPLEX FRAMES, EXACT LABELS, AND QUADRATIC EVALUATIONS

Put

$$q_i = e_i - \frac{1}{N}\mathbf{1}, \quad \Pi = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^T.$$

For $A \in \mathcal{O}_d$, write $p_i = Aq_i$.

Lemma 2.1. *The vectors p_i satisfy*

$$\sum_i p_i = 0, \quad \langle p_i, p_j \rangle = \delta_{ij} - \frac{1}{N}, \quad \sum_i p_i p_i^T = I_d. \quad (2.1)$$

Consequently, the d coordinate columns of the $N \times d$ matrix with rows p_i^T are orthonormal and perpendicular to $\mathbf{1}$.

Proof. The first two assertions follow by expanding $q_i = e_i - N^{-1}\mathbf{1}$. Also $\sum_i q_i q_i^T = \Pi$, whose restriction to V is the identity. Apply the orthogonal map A to obtain (2.1). The last assertion is the matrix form of the first and third identities. $\square$

For an odd function G on the unit sphere, define the skew-symmetric label matrix $Y_G(A)$ by

$$Y_G(A)_{ij} = G\left(\frac{p_i - p_j}{\sqrt{2}}\right) \quad (i \neq j), \quad Y_G(A)_{ii} = 0. \quad (2.2)$$

For polynomials the same notation refers to their restrictions to the sphere. Define the residual

$$\mathcal{R}_G(A) = \Pi Y_G(A) \Pi. \quad (2.3)$$

Lemma 2.2 (Exact labels). *For a skew-symmetric real $N \times N$ matrix Y , the following are equivalent:*

- (i) $Y_{ij} = \langle c, \alpha_{ij} \rangle$ for all $i < j$, for some $c \in V$;
- (ii) $Y_{ij} = \rho_i - \rho_j$ for real numbers $\rho_0, \dots, \rho_d$;
- (iii) $\Pi Y \Pi = 0$.

Thus (1.3) at a specified A is equivalent to $\mathcal{R}_F(A) = 0$.

Proof. Condition (i) gives (ii) with $\rho_i = c_i/\sqrt{2}$. Conversely, subtract the mean from the ρ_i and take $c_i = \sqrt{2}\rho_i$. This gives $c \in V$ and (i).

Under (ii), $Y = \rho \mathbf{1}^T - \mathbf{1} \rho^T$, so (iii) follows from $\Pi \mathbf{1} = 0$. For the converse put $J_0 = N^{-1} \mathbf{1} \mathbf{1}^T$. Skew-symmetry gives $J_0 Y J_0 = 0$, and expanding (iii) yields $Y = J_0 Y + Y J_0$. Its entries are $\rho_i - \rho_j$, where $\rho_i = N^{-1} \sum_j Y_{ij}$. Finally, in (1.3) one has $c = A^T b$, and every $c \in V$ arises from $b = Ac$. $\square$

Matrices satisfying these conditions will be called *exact*. If X is skew-symmetric and $X \mathbf{1} = 0$, then

$$\sum_{i < j} X_{ij}(\rho_i - \rho_j) = \sum_i \rho_i \sum_j X_{ij} = 0. \quad (2.4)$$

This is the cut/cycle orthogonality used later.

Lemma 2.3 (Quadratic evaluation isomorphism). *The linear map*

$$\text{Sym}(V) \longrightarrow \mathbb{R}^n, \quad B \longmapsto (\langle \alpha_{ij}, B \alpha_{ij} \rangle)_{i < j} \quad (2.5)$$

is an isomorphism.

Proof. Extend B to $\mathbb{R}^N$ by setting $B \mathbf{1} = 0$. Form the symmetric matrix W with zero diagonal and entries

$$W_{ij} = \frac{B_{ii} + B_{jj}}{2} - B_{ij} = \langle \alpha_{ij}, B \alpha_{ij} \rangle \quad (i \neq j).$$

The same displayed expression is zero when $i = j$. Hence $\Pi W \Pi = -B$, since $\Pi B \Pi = B$ and the terms involving B_{ii} or B_{jj} are killed by the projections. Thus (2.5) is injective. Both spaces have dimension $d(d+1)/2 = n$, proving surjectivity as well. $\square$

3. SEMIALGEBRAIC DIMENSION FACTS

We use real semialgebraic dimension, with $\dim \emptyset = -\infty$. The needed foundational results are the Tarski–Seidenberg theorem, semialgebraic dimension theory, and Hardt’s triviality theorem; see [BCR98, Sections 2.2, 2.8, and 9.3] and [H80]. An accessible account of the stratification and trivialization statements is [HLTV11, Section 2.1].

Lemma 3.1. *Semialgebraic images are semialgebraic. Finite unions have dimension equal to the maximum of the dimensions, and dimensions add under Cartesian products. A semialgebraic subset of $\mathbb{R}^s$ of dimension less than s has empty interior.*

Let $\pi : X \rightarrow Y$ be a continuous semialgebraic map. If every nonempty fiber has dimension at most k , then

$$\dim X \leq \dim \pi(X) + k. \quad (3.1)$$

If every nonempty fiber has dimension at least k , then

$$\dim X \geq \dim \pi(X) + k. \quad (3.2)$$

In particular, when all nonempty fibers have dimension k , equality holds.

Proof. The first assertions are the cited basic dimension results. By semialgebraic triviality, $\pi(X)$ has a finite semialgebraic partition Y_a over which $\pi^{-1}(Y_a)$ is semialgebraically homeomorphic to $Y_a \times F_a$, with π corresponding to projection. The two fiber inequalities follow from

$$\dim X = \max_a (\dim Y_a + \dim F_a), \quad \dim \pi(X) = \max_a \dim Y_a. \quad \square$$

4. GENERIC AVOIDANCE OUTSIDE THE EXCEPTIONAL SET

4.1. Orbit separation and interpolation. The action (1.5) is semialgebraic: write $\cos \theta = a$, $\sin \theta = b$ with $a^2 + b^2 = 1$, and use $\cos 2\theta = a^2 - b^2$, $\sin 2\theta = 2ab$.

Definition 4.1. A placement $A \in \mathcal{O}_d$ is *exceptional* if $A\alpha_{ij}$ belongs to $E_1 \cup E_2$ for at least one $i < j$. Write $\mathcal{E}$ for the set of exceptional placements.

The set $\mathcal{E}$ is a finite union of closed subsets of the compact space $\mathcal{O}_d$, and hence is compact.

Lemma 4.2. *For $w \in S^{d-1}$, the orbit of w under (1.5) contains $-w$ if and only if $w \in E_1 \cup E_2$. Moreover, the polynomial map*

$$\Psi(z_1, z_2, t) = (t, |z_1|^2, |z_2|^2, \operatorname{Re}(z_1^2 \bar{z}_2), \operatorname{Im}(z_1^2 \bar{z}_2)) \quad (4.1)$$

separates circle orbits in $\mathbb{R}^d$ and has degree at most three.

Proof. If $h_\theta w = -w$, the fixed component t vanishes. If both complex coordinates were nonzero, we would have simultaneously $e^{i\theta} = -1$ and $e^{2i\theta} = -1$, which is impossible. Conversely, $\theta = \pi$ reverses a vector of E_1 and $\theta = \pi/2$ reverses a vector of E_2 .

The entries of Ψ are invariant. Suppose two points have the same image. If their first complex coordinates are nonzero, their equal moduli permit a unique phase rotation taking the first coordinate of one point to that of the other. Equality of $z_1^2 \bar{z}_2$ then gives equality of the second coordinates after that same rotation. If both first coordinates vanish, equality of $|z_2|$ suffices, since multiplication by $e^{2i\theta}$ is transitive on each circle. The fixed components are explicitly recorded. This also covers the case of two zero complex coordinates. $\square$

Lemma 4.3 (Invariant interpolation). *Let $w_1, \dots, w_m \in S^{d-1}$, where $m \leq n$, be such that the $2m$ circle orbits of $w_1, -w_1, \dots, w_m, -w_m$ are distinct. Then the evaluation map*

$$\mathcal{P}_d \longrightarrow \mathbb{R}^m, \quad F \longmapsto (F(w_1), \dots, F(w_m))$$

is surjective.

Proof. Let $\xi_1, \dots, \xi_{2m}$ be the distinct points obtained by applying Ψ to the indicated signed points. For each j , define a polynomial in the target coordinates Y by

$$L_j(Y) = \prod_{k \neq j} \frac{\langle Y - \xi_k, \xi_j - \xi_k \rangle}{\|\xi_j - \xi_k\|^2}.$$

It has degree at most $2m - 1$ and satisfies $L_j(\xi_k) = \delta_{jk}$. For arbitrary prescribed real values $a_1, \dots, a_m$, a linear combination P of these polynomials takes the value a_j at $\Psi(w_j)$ and $-a_j$ at $\Psi(-w_j)$. The polynomial $P \circ \Psi$ is circle-invariant and has degree at most $3(2m - 1) \leq D$. Its odd part

$$F(w) = \frac{P(\Psi(w)) - P(\Psi(-w))}{2}$$

is still invariant, belongs to $\mathcal{P}_d$, and has $F(w_j) = a_j$. $\square$

4.2. A rank-one spectral fiber bound. For $\lambda = (\lambda_1, \lambda_2) \in (0, 1) \times (4, 5)$ set

$$D_\lambda = \lambda_1 I_{E_1} \oplus \lambda_2 I_{E_2} \oplus \operatorname{diag}(\mu_1, \dots, \mu_{d-4}), \quad \mu_j = 4j + 4. \quad (4.2)$$

The final block is absent when $d = 4$. Distinct eigenvalues of D_λ are separated by more than three.

Lemma 4.4 (Spectral fibers). *Every fiber of the semialgebraic map*

$$\Phi : \mathcal{O}_d \times (0, 1) \times (4, 5) \times \{c \in V : \|c\| < 1\} \longrightarrow \operatorname{Sym}(V), \quad \Phi(A, \lambda_1, \lambda_2, c) = A^T D_\lambda A + cc^T \quad (4.3)$$

has dimension at most two.

Proof. Fix a matrix B in the image and write

$$S = A^T D_\lambda A, \quad B = S + cc^T.$$

In each two-dimensional eigenspace of S for λ_1 or λ_2 , there is a nonzero vector perpendicular to c . It remains an eigenvector of B with the same eigenvalue. Consequently $\lambda_1, \lambda_2 \in \text{spec } B$, leaving only finitely many possibilities for the pair λ .

Fix such a pair. Write $s_1 \leq \dots \leq s_d$ for the eigenvalues of S and $\beta_1 \leq \dots \leq \beta_d$ for those of B , with multiplicities. The inequalities

$$S \leq B \leq S + \|c\|^2 I$$

in the order of symmetric matrices imply

$$s_j \leq \beta_j \leq s_j + \|c\|^2 < s_j + 1. \quad (4.4)$$

For completeness, this follows from

$$\lambda_j(T) = \min_{\dim W=j} \max_{\substack{x \in W \\ \|x\|=1}} \langle x, Tx \rangle.$$

The formula itself follows from an orthonormal eigenbasis: the span of the first j eigenvectors gives one inequality, and every j -dimensional subspace meets the span of eigenvectors numbered $j, \dots, d$ nontrivially, giving the other.

The intervals $[s, s+1)$ attached to distinct eigenvalues s of S are disjoint. A simple eigenvalue of S contributes only one ordered eigenvalue of B to its interval. A double eigenvalue contributes two, and at least one equals the original eigenvalue. It follows that an eigenvalue of B has multiplicity at most two, and any double eigenvalue of B is either λ_1 or λ_2 .

Let P_β be the orthogonal projection to the β -eigenspace of B . The determinant identity $\det(I + uv^T) = 1 + v^T u$, obtained by multilinearity in the columns, gives the rational-function identity

$$\frac{\det(tI - S)}{\det(tI - B)} = 1 + c^T(tI - B)^{-1}c = 1 + \sum_{\beta \in \text{spec } B} \frac{\|P_\beta c\|^2}{t - \beta}. \quad (4.5)$$

The left side is determined by B and λ . Its residues therefore determine every $\|P_\beta c\|^2$. If β is double in B , it is also double in S , so the left side has no pole at β . Thus $P_\beta c = 0$. On every remaining, one-dimensional eigenspace, the component of c is determined up to sign. There are only finitely many possible c .

For each such choice, $S = B - cc^T$ is fixed. The set of A solving $A^T D_\lambda A = S$ is a coset of the orthogonal centralizer of D_λ , which is

$$O(2) \times O(2) \times \{\pm 1\}^{d-4}.$$

Indeed, a commuting orthogonal map preserves each distinct eigenspace, and any orthogonal map on each eigenspace commutes. Each $O(2)$ has dimension one, so this centralizer has dimension two. The fiber is contained in a finite union of such cosets, proving the claim. $\square$

4.3. Signed orbit strata. Index positive roots by edges $e = (i, j)$ with $i < j$. For $A \notin \mathcal{E}$, put two edges in the same class when

$$A\alpha_e = \sigma h_\theta A\alpha_{e'} \quad \text{for some } \sigma \in \{\pm 1\}, \theta \in \mathbb{R}.$$

By lemma 4.2, the sign relative to a fixed representative is unique. Choose the least edge in each class as its representative, and give that representative sign $+1$. There are finitely many resulting signed partitions.

Fix one signed partition, with m classes and representatives $e_1, \dots, e_m$, and let $Z \subset \mathcal{O}_d \setminus \mathcal{E}$ be the set of placements realizing it exactly. It is semialgebraic: relations and nonrelations are expressed

using equalities and inequalities between values of (4.1). It is also preserved by left multiplication by every h_θ . Define

$$L_Z = \{c \in V : \langle c, \alpha_e \rangle = \sigma_e \langle c, \alpha_{e_a} \rangle \text{ for } e \text{ in class } a\}, \quad (4.6)$$

$$\mathcal{S}_Z = \{B \in \text{Sym}(V) : \langle \alpha_e, B\alpha_e \rangle = \langle \alpha_{e_a}, B\alpha_{e_a} \rangle \text{ for } e \text{ in class } a\}. \quad (4.7)$$

Write $\ell = \dim L_Z$. By lemma 2.3,

$$\dim \mathcal{S}_Z = m. \quad (4.8)$$

Lemma 4.5 (Dimension of a signed stratum). *For every nonempty signed stratum Z ,*

$$\dim Z + \ell \leq m. \quad (4.9)$$

Proof. For $A \in Z$, $c \in L_Z$, and the parameters in (4.2), we have

$$A^T D_\lambda A + cc^T \in \mathcal{S}_Z. \quad (4.10)$$

For the first term, this is because D_λ commutes with h_θ and quadratic evaluations do not see the sign σ_e . For the second term, it follows by squaring the defining equalities of L_Z .

Restrict (4.3) to

$$Z \times (0, 1) \times (4, 5) \times (L_Z \cap \{\|c\| < 1\}).$$

The domain has dimension $\dim Z + 2 + \ell$, and its image lies in the m -dimensional space (4.7). Its fibers have dimension at most two by lemma 4.4. Applying (3.1) yields $\dim Z + 2 + \ell \leq m + 2$, which is (4.9). The restriction $\|c\| < 1$ is used only for this auxiliary spectral argument; it places no bound on the eventual linear functional in the agreement problem. $\square$

Proposition 4.6 (Generic nonexceptional avoidance). *The set*

$$\mathcal{B} = \{F \in \mathcal{P}_d : \mathcal{R}_F(A) = 0 \text{ for some } A \notin \mathcal{E}\}$$

is semialgebraic and has dimension at most $\dim \mathcal{P}_d - 1$. In particular, $\mathcal{P}_d \setminus \mathcal{B}$ is dense in $\mathcal{P}_d$.

Proof. Fix a nonempty stratum Z as above, and put $p = \dim \mathcal{P}_d$. For each $A \in Z$, the orbits of the representatives and their negatives are all distinct. Thus lemma 4.3 makes the m -value evaluation map surjective. Consider the semialgebraic incidence set

$$\mathcal{I}_Z = \{(A, c, F) \in Z \times L_Z \times \mathcal{P}_d : F(A\alpha_{e_a}) = \langle c, \alpha_{e_a} \rangle \ (1 \leq a \leq m)\}. \quad (4.11)$$

Over every $(A, c) \in Z \times L_Z$, its fiber is a nonempty affine space of dimension $p - m$. By lemmas 3.1 and 4.5,

$$\dim \mathcal{I}_Z = \dim Z + \ell + p - m \leq p. \quad (4.12)$$

Every nonempty fiber of the projection $\mathcal{I}_Z \rightarrow \mathcal{P}_d$ contains the circle

$$\{(h_\theta A, c, F) : \theta \in \mathbb{R}/2\pi\mathbb{Z}\}.$$

It lies in the same fiber because F is invariant and Z is preserved by the action. It is one-dimensional: if $h_\theta A = A$, the surjectivity of A gives $h_\theta = I$, and its action on E_1 gives $\theta = 0$ modulo 2π . The circle is semialgebraic by the polynomial parametrization preceding definition 4.1. Consequently (3.2) and (4.12) bound the projection's dimension by $p - 1$.

A polynomial has exact labels at a placement in Z if and only if it occurs in this projection. For the forward implication, lemma 2.2 supplies c , and signed orbit invariance forces $c \in L_Z$. For the converse, the representative equations in (4.11) extend to every edge by oddness and circle invariance. Taking the finite union of these projections over all signed partitions gives precisely $\mathcal{B}$. The final assertion follows because a smaller-dimensional semialgebraic set has empty interior. $\square$

5. THE CUBIC IDENTITIES

Write $z_1 = x + iy$, $z_2 = u + iv$. Then

$$Q = (x^2 - y^2)u + 2xyv, \quad H = (x^2 + y^2)(2xyu - (x^2 - y^2)v). \quad (5.1)$$

Let

$$J(z_1, z_2, t) = (iz_1, 2iz_2, 0), \quad X(A)_{ij} = \langle p_i, Jp_j \rangle.$$

For an odd function G define

$$\Lambda_A(G) = \sum_{i < j} X(A)_{ij} G\left(\frac{p_i - p_j}{\sqrt{2}}\right). \quad (5.2)$$

The matrix $X(A)$ is skew-symmetric and $X(A)\mathbf{1} = 0$ by (2.1). Hence (2.4) gives

$$\mathcal{R}_G(A) = 0 \implies \Lambda_A(G) = 0. \quad (5.3)$$

The seed and the annihilator used in this section are those of [CG26]. We give direct proofs, including the normalization constants, because the later argument uses them in every dimension.

Lemma 5.1 (Cubic expansion). *For any real homogeneous cubic C and vectors a, b ,*

$$C(a - b) = C(a) - C(b) - \langle \nabla C(a), b \rangle + \langle a, \nabla C(b) \rangle. \quad (5.4)$$

Proof. Write $C(w) = T(w, w, w)$ for its symmetric trilinear polarization. Expansion of $T(a - b, a - b, a - b)$ gives mixed terms $-3T(a, a, b) + 3T(a, b, b)$, which are exactly the two gradient terms in (5.4). $\square$

Lemma 5.2 (Cubic annihilation). *For every $A \in \mathcal{O}_d$,*

$$\Lambda_A(Q) = 0. \quad (5.5)$$

Proof. Apply (5.4) to (5.2). The terms $Q(p_i) - Q(p_j)$ are exact and make no contribution. Also, by (2.1) and the skew-symmetry of J ,

$$\sum_j X(A)_{ij} p_j = -Jp_i, \quad \sum_i X(A)_{ij} p_i = Jp_j.$$

Since the product of two skew-symmetric edge labels is unchanged when i, j are exchanged, the sum over $i < j$ is half the sum over all ordered pairs. Therefore

$$\begin{aligned} \Lambda_A(Q) &= \frac{1}{4\sqrt{2}} \sum_{i,j} X(A)_{ij} (-\langle \nabla Q(p_i), p_j \rangle + \langle p_i, \nabla Q(p_j) \rangle) \\ &= \frac{1}{2\sqrt{2}} \sum_i \langle \nabla Q(p_i), Jp_i \rangle = 0. \end{aligned}$$

The last equality follows by differentiating $Q(h_\theta w) = Q(w)$ at $\theta = 0$. $\square$

Let $x, y, u, v \in \mathbb{R}^N$ now denote the first four coordinate *columns* of the simplex frame (p_i) . All products of such columns below are componentwise. Set

$$P = (x \ y \ u \ v), \quad G = (2xu + 2yv \quad -2yu + 2xv \quad x^2 - y^2 \quad 2xy). \quad (5.6)$$

Thus the rows of G are the first four gradient coordinates of Q .

Lemma 5.3 (Symmetric gradient matrix). *If $\mathcal{R}_Q(A) = 0$, there is a symmetric real 4×4 matrix S such that*

$$G = PS. \quad (5.7)$$

Proof. By lemma 2.1, $P^T P = I_4$ and $\mathbf{1}^T P = 0$. The same identities give $\mathbf{1}^T G = 0$: the mixed products have zero sum, and $\sum_i (x_i^2 - y_i^2) = 1 - 1 = 0$. Applying (5.4) to every edge and projecting away the exact term gives

$$\mathcal{R}_Q(A) = \frac{1}{2\sqrt{2}}(PG^T - GP^T). \quad (5.8)$$

Exactness therefore gives $PG^T = GP^T$. Multiplication on the right by P yields $G = P(G^T P)$, while multiplication on both sides by P^T and P gives $G^T P = P^T G$. Thus $S = G^T P$ is symmetric and satisfies (5.7). $\square$

Lemma 5.4 (Allowed normalizations). *Each of the following operations preserves exactness of the Q labels and preserves the number $\Lambda_A(H)$:*

- (i) left multiplication of A by h_θ ;
- (ii) simultaneous complex conjugation $C(z_1, z_2, t) = (\bar{z}_1, \bar{z}_2, t)$;
- (iii) a permutation of the simplex indices.

Proof. The maps h_θ preserve Q, H and commute with J . For conjugation, $Q(Cw) = Q(w)$, $H(Cw) = -H(w)$, and $C^T J C = -J$. Thus both factors of each summand of $\Lambda_A(H)$ change sign. Finally, the exactness condition is invariant under relabeling. The identity

$$\Lambda_A(H) = \frac{1}{2} \sum_{i,j} X(A)_{ij} Y_H(A)_{ij}$$

shows invariance under permutations as well. A permutation of indices is realized by the corresponding orthogonal automorphism of V , so all the resulting frames are still legitimate placements. $\square$

6. EXCEPTIONAL CUBIC SOLUTIONS AND THE QUINTIC OBSTRUCTION

The argument in this section classifies only the information needed at exceptional cubic solutions. It does *not* classify every solution of $\mathcal{R}_Q(A) = 0$.

Proposition 6.1 (Necessary normal form). *Suppose $A \in \mathcal{E}$ and $\mathcal{R}_Q(A) = 0$. After operations allowed in lemma 5.4, there are disjoint sets of indices K_1, K_2 , of even cardinalities $k_1, k_2 \geq 2$, and real numbers $\gamma \neq 0$ and ω such that*

$$x_i = \begin{cases} \pm k_1^{-1/2}, & i \in K_1, \\ 0, & i \notin K_1, \end{cases} \quad y_i = \begin{cases} \pm k_2^{-1/2}, & i \in K_2, \\ 0, & i \notin K_2. \end{cases} \quad (6.1)$$

On each support the plus and minus signs occur equally often, and

$$u = \frac{x^2 - y^2}{\gamma}, \quad v_i = \omega \quad (i \in K_1 \cup K_2), \quad \gamma^2 = \frac{1}{k_1} + \frac{1}{k_2}. \quad (6.2)$$

Proof. Choose a root in $E_1 \cup E_2$ and distinguish the two possibilities. Throughout the proof, the coordinate columns are orthonormal, have zero mean, and satisfy (5.7).

Case 1: a root belongs to E_1 . A circle rotation and a relabeling make this root the real z_1 axis and its indices 0, 1. Since $A^T e_x = \alpha_{01}$, the column x is

$$x_0 = \frac{1}{\sqrt{2}}, \quad x_1 = -\frac{1}{\sqrt{2}}, \quad x_i = 0 \quad (i \neq 0, 1).$$

The other coordinate columns have equal entries at indices 0, 1. Write

$$y_0 = y_1 = \tau, \quad u_0 = u_1 = U, \quad v_0 = v_1 = W.$$

Subtracting rows 0 and 1 in (5.7) gives

$$(S_{xx}, S_{xy}, S_{xu}, S_{xv}) = (2U, 2W, 0, 2\tau). \quad (6.3)$$

Here the subscripts indicate the ordered coordinates x, y, u, v . Since $2xy = 2\tau x$, the last column of G and the independence of the columns of P give

$$S_{xv} = 2\tau, \quad S_{yv} = S_{uv} = S_{vv} = 0.$$

Symmetry of S and the remaining columns of (5.7) consequently give

$$yv = Wy + \tau v, \quad (6.4)$$

$$yu = ay + bu, \quad (6.5)$$

$$x^2 - y^2 = -2by + \gamma u, \quad (6.6)$$

for some real a, b, γ . Specifically, $a = -S_{yy}/2$, $b = -S_{yu}/2$, and $\gamma = S_{uu}$. Taking the inner product of (6.6) with u and using (6.5) gives

$$\gamma = \langle x^2, u \rangle - \langle y^2, u \rangle = U - \langle y, yu \rangle = U - a. \quad (6.7)$$

This derives all coefficients used below.

We first prove $\tau = 0$. At index 0, (6.4) implies $\tau W = 0$. If $\tau \neq 0$, then $W = 0$ and $(y - \tau \mathbf{1})v = 0$. Thus v is supported where $y_i = \tau$. The unit column v vanishes at 0, 1, so there exists an index $i \notin \{0, 1\}$ with $v_i \neq 0$ and $y_i = \tau$. Comparing (6.6) at 0 and i gives

$$\gamma(U - u_i) = \frac{1}{2}.$$

In particular $U \neq u_i$. Comparison of (6.5) at these indices gives $(\tau - b)(U - u_i) = 0$, so $b = \tau$. Substitution at index 0 then gives $a = 0$. Equation (6.5) becomes $yu = \tau u$. For an index $j \notin \{0, 1\}$ with $y_j \neq \tau$, this forces $u_j = 0$, and (6.6) gives $-y_j^2 = -2\tau y_j$. Therefore every entry of y lies in $\{0, \tau, 2\tau\}$. Since $y_0 = \tau \neq 0$, all nonzero entries have the same sign, contradicting $\sum_j y_j = 0$. This proves $\tau = 0$.

At index 0, (6.6) now gives $\gamma U = 1/2$, so $\gamma, U \neq 0$. Equation (6.5) at that index gives $b = 0$. Hence

$$u = \frac{x^2 - y^2}{\gamma}, \quad yu = ay. \quad (6.8)$$

The supports of x and y are disjoint because $y_0 = y_1 = 0$. On the nonempty support of y , the second equation says $u_i = a$; the first says $y_i^2 = -\gamma a$. This is a fixed positive value. Because y has zero mean and norm one, its support has even cardinality and its nonzero entries have equal positive and negative multiplicities, with magnitude the reciprocal square root of that cardinality. Equation (6.4) now says $v_i = W$ on this support; the same holds on $\{0, 1\}$ by definition. This is (6.1)–(6.2), with $K_1 = \{0, 1\}$ and $\omega = W$, apart from the final norm identity, which is proved after Case 2.

Case 2: a root belongs to E_2 . A circle rotation and relabeling make the root the real z_2 axis, with indices 0, 1. Thus

$$u_0 = \frac{1}{\sqrt{2}}, \quad u_1 = -\frac{1}{\sqrt{2}}, \quad u_i = 0 \quad (i \neq 0, 1).$$

Write $x_0 = x_1 = X$ and $y_0 = y_1 = Y$. Subtracting rows in (5.7) gives the u row of S as $(2X, -2Y, 0, 0)$. Symmetry therefore gives

$$x^2 - y^2 = 2Xx - 2Yy.$$

At index 0, this implies $X^2 = Y^2$. If necessary, simultaneous complex conjugation changes Y to $-Y$, so assume $X = Y$. Put

$$r = \frac{x - y}{\sqrt{2}}, \quad s = \frac{x + y}{\sqrt{2}}, \quad \eta = \sqrt{2}X. \quad (6.9)$$

These are orthonormal columns and satisfy

$$rs = \eta r, \quad ru = 0, \quad su = \eta u, \quad r_0 = r_1 = 0, \quad s_0 = s_1 = \eta. \quad (6.10)$$

In the orthogonal coordinates r, s, u, v the cubic becomes

$$Q = 2rsu + (s^2 - r^2)v,$$

and its gradient columns become

$$(2su - 2rv, \quad 2ru + 2sv, \quad 2rs, \quad s^2 - r^2).$$

The relation (5.7) remains valid with a symmetric matrix after any orthogonal change of these four coordinates. By (6.10), its third gradient column is $2\eta r$. Symmetry fixes the u coefficients in the other columns. The additional identities

$$\langle rv, s \rangle = \langle sv, r \rangle = \sum_i r_i s_i v_i = \eta \langle r, v \rangle = 0$$

remove their cross coefficients. It follows that there are real a, c, e, f, γ with

$$rv = ar + cv, \quad sv = es + fv, \quad (6.11)$$

$$s^2 - r^2 = -2cr + 2fs + \gamma v. \quad (6.12)$$

The coefficients $-2c$ and $2f$ in the last equation follow from symmetry, and inner product with v gives

$$\gamma = \langle s, sv \rangle - \langle r, rv \rangle = e - a. \quad (6.13)$$

We claim $\eta = 0$. Suppose otherwise, and let $K = \text{supp } r$. It is nonempty, and $s_i = \eta$ for $i \in K$. If $c \neq 0$, the first equation of (6.11) forces $v_i = 0$ off K . Thus $sv = \eta v$, so independence of s, v in the second equation gives $e = 0$ and $f = \eta$. Off K , (6.12) now gives $s_i^2 = 2\eta s_i$. On K , one has $s_i = \eta$. All entries of s therefore belong to $\{0, \eta, 2\eta\}$, contradicting its zero mean. Hence $c = 0$.

The first equation of (6.11) now gives $v_i = a$ on K . Let $v_0 = v_1 = W$. Comparing (6.12) at 0 and an index $i \in K$ yields

$$r_i^2 = \gamma(W - a) > 0.$$

In particular $W \neq a$. Comparing the second equation of (6.11) at those same indices gives $(\eta - f)(W - a) = 0$. Therefore $f = \eta$, and substitution gives $e = 0$. Now $(s - \eta \mathbf{1})v = 0$. At any index outside K where $s_i \neq \eta$, one has $v_i = 0$, and (6.12) yields $s_i \in \{0, 2\eta\}$. All other entries equal η . This gives the same zero-mean contradiction. Thus $\eta = 0$.

We now have $rs = 0$. Multiplying the first equation of (6.11) by s gives $ce = cf = 0$; multiplying the second by r gives $fa = fc = 0$, since the relevant columns are linearly independent. If $c \neq 0$, then $e = f = 0$, so $sv = 0$. At an index where $s_i \neq 0$, both r_i and v_i vanish, and (6.12) would give $s_i^2 = 0$, a contradiction. Hence $c = 0$. Similarly, if $f \neq 0$, then $a = c = 0$ and $rv = 0$; at an index where $r_i \neq 0$, the same equation gives $-r_i^2 = 0$. Therefore $f = 0$ as well. We have proved

$$rv = ar, \quad sv = es, \quad s^2 - r^2 = \gamma v. \quad (6.14)$$

The number γ is nonzero: otherwise $r^2 = s^2$, impossible for nonzero columns with disjoint supports.

On the support of r one has $v_i = a$ and $r_i^2 = -\gamma a$; on the support of s one has $v_i = e$ and $s_i^2 = \gamma e$. Thus each nonzero magnitude is constant on its support. Zero mean and unit norm give even support sizes and balanced signs as in (6.1). Outside the two supports, $v_i = 0$.

Finally apply the allowed circle rotation $h_{\pi/4}$. By (6.9), the new complex columns are

$$z'_1 = r + is, \quad z'_2 = -v + iu.$$

Thus the new real z_2 column is $(r^2 - s^2)/\gamma$. The new imaginary column is the old u , which is supported at indices 0, 1, where $r = s = 0$. Consequently it is zero on both supports. This proves the asserted form with $\omega = 0$.

The final norm identity. In either case, disjointness of the supports and the fact that u has norm one give

$$1 = \|u\|^2 = \frac{1}{\gamma^2} \left(\sum_i x_i^4 + \sum_i y_i^4 \right) = \frac{1}{\gamma^2} \left(\frac{1}{k_1} + \frac{1}{k_2} \right).$$

This completes the proof of (6.2) and the proposition. $\square$

Lemma 6.2 (Evaluation in the normal form). *For a frame in the form of proposition 6.1,*

$$\Lambda_A(H) = \frac{1}{\sqrt{2} k_1 k_2 \gamma} \neq 0. \quad (6.15)$$

Proof. For an oriented edge (i, j) write $\Delta x = x_i - x_j$ and similarly for the other coordinates. The normalization by $\sqrt{2}$ gives

$$H\left(\frac{p_i - p_j}{\sqrt{2}}\right) = \frac{(\Delta x^2 + \Delta y^2)(2\Delta x \Delta y \Delta u - (\Delta x^2 - \Delta y^2)\Delta v)}{4\sqrt{2}}. \quad (6.16)$$

Here Δx^2 means $(\Delta x)^2$, and likewise for Δy . The generator gives

$$X(A)_{ij} = -x_i y_j + y_i x_j - 2u_i v_j + 2v_i u_j. \quad (6.17)$$

Let O be the complement of $K_1 \cup K_2$, and put

$$U_1 = \frac{1}{k_1 \gamma}, \quad U_2 = -\frac{1}{k_2 \gamma}.$$

Then $u_i = U_a$ on K_a , $u_i = 0$ on O , and $U_1 - U_2 = \gamma$ by (6.2). We group the edge contributions. The product of the two factors in (5.2) is independent of edge orientation, so each edge may be oriented to agree with the indicated grouping.

Edges between K_1 and K_2 . For $i \in K_1$, $j \in K_2$, put $a = x_i$ and $b = y_j$. Equations (6.16) and (6.17) give

$$H\left(\frac{p_i - p_j}{\sqrt{2}}\right) = -\frac{\gamma^3 ab}{2\sqrt{2}}, \quad X(A)_{ij} = -ab - 2\omega\gamma.$$

Consequently the sum over these edges is

$$\frac{\gamma^3}{2\sqrt{2}} \sum_{i \in K_1, j \in K_2} (x_i^2 y_j^2 + 2\omega\gamma x_i y_j) = \frac{\gamma^3}{2\sqrt{2}}. \quad (6.18)$$

We used $\sum_{K_1} x_i^2 = \sum_{K_2} y_j^2 = 1$ and $\sum_{K_1} x_i = \sum_{K_2} y_j = 0$.

Edges from K_1 to O . Fix $j \in O$ and write $w = v_j$. For $i \in K_1$,

$$H\left(\frac{p_i - p_j}{\sqrt{2}}\right) = -\frac{\omega - w}{4\sqrt{2} k_1^2}, \quad X(A)_{ij} = -2U_1 w.$$

Summing over the k_1 possible indices i gives

$$\frac{w(\omega - w)}{2\sqrt{2} k_1^2 \gamma}. \quad (6.19)$$

Edges from K_2 to O . For the same fixed j and $i \in K_2$,

$$H\left(\frac{p_i - p_j}{\sqrt{2}}\right) = \frac{\omega - w}{4\sqrt{2} k_2^2}, \quad X(A)_{ij} = -2U_2 w.$$

Their sum is

$$\frac{w(\omega - w)}{2\sqrt{2} k_2^2 \gamma}. \quad (6.20)$$

All remaining edges and the total. Edges internal to K_1 or to K_2 have $\Delta u = \Delta v = 0$, so their H labels vanish. Edges internal to O have $\Delta x = \Delta y = 0$, so their labels also vanish. Because v has zero mean and norm one,

$$\sum_{j \in O} v_j = -(k_1 + k_2)\omega, \quad \sum_{j \in O} v_j^2 = 1 - (k_1 + k_2)\omega^2.$$

It follows that

$$\sum_{j \in O} v_j(\omega - v_j) = -1. \quad (6.21)$$

Thus (6.18)–(6.21) give

$$\begin{aligned} \Lambda_A(H) &= \frac{1}{2\sqrt{2}} \left(\gamma^3 - \frac{k_1^{-2} + k_2^{-2}}{\gamma} \right) \\ &= \frac{(k_1^{-1} + k_2^{-1})^2 - k_1^{-2} - k_2^{-2}}{2\sqrt{2}\gamma} = \frac{1}{\sqrt{2}k_1k_2\gamma}. \end{aligned}$$

The denominator is nonzero. This proves the assertion. $\square$

Corollary 6.3 (Exceptional obstruction). *If $A \in \mathcal{E}$ and $\mathcal{R}_Q(A) = 0$, then $\Lambda_A(H) \neq 0$.*

Proof. Apply proposition 6.1 and lemma 6.2, and use the invariance of $\Lambda_A(H)$ under the normalizations in lemma 5.4. $\square$

7. COMPLETION OF THE COUNTEREXAMPLE CONSTRUCTION

Proposition 7.1 (An open set avoiding exceptional placements). *There exists $\varepsilon_0 > 0$ such that, for every $0 < |\varepsilon| < \varepsilon_0$, the polynomial $F_\varepsilon = Q + \varepsilon H$ satisfies*

$$\mathcal{R}_{F_\varepsilon}(A) \neq 0 \quad (A \in \mathcal{E}).$$

For each such ε , this property persists in a neighborhood of F_ε in $\mathcal{P}_d$.

Proof. If the first assertion failed, there would exist nonzero $\varepsilon_j \rightarrow 0$ and $A_j \in \mathcal{E}$ such that $\mathcal{R}_{Q+\varepsilon_j H}(A_j) = 0$. By (5.3) and lemma 5.2,

$$0 = \Lambda_{A_j}(Q + \varepsilon_j H) = \varepsilon_j \Lambda_{A_j}(H).$$

Hence $\Lambda_{A_j}(H) = 0$ for every j . Compactness gives a subsequence converging to $A_\infty \in \mathcal{E}$. The residual is continuous, so $\mathcal{R}_Q(A_\infty) = 0$; likewise, $\Lambda_{A_\infty}(H) = 0$. This contradicts corollary 6.3.

Fix an admissible ε . Continuity and compactness give

$$\delta = \min_{A \in \mathcal{E}} \|\mathcal{R}_{F_\varepsilon}(A)\|_F > 0, \quad (7.1)$$

where $\|\cdot\|_F$ is the Frobenius matrix norm. Because $\mathcal{P}_d$ is finite-dimensional and $\mathcal{O}_d$ is compact, there is a constant $C > 0$ such that

$$\sup_{A \in \mathcal{O}_d} \|\mathcal{R}_R(A)\|_F \leq C\|R\| \quad (R \in \mathcal{P}_d). \quad (7.2)$$

For example, expand R in a fixed basis of $\mathcal{P}_d$, bound each basis residual on $\mathcal{O}_d$, and use equivalence of norms in a finite-dimensional vector space. By linearity of the residual, every $F_\varepsilon + R$ with $\|R\| < \delta/(2C)$ has residual norm at least $\delta/2$ on $\mathcal{E}$. This proves the openness assertion. $\square$

Proof of Theorems 1.1 and 1.2. Fix $0 < |\varepsilon| < \varepsilon_0$ as in proposition 7.1, and let $\eta > 0$. By proposition 4.6, the complement of $\mathcal{B}$ is dense in $\mathcal{P}_d$. The open ball about F_ε of radius $\min\{\eta, \delta/(2C)\}$ therefore contains a polynomial $F \notin \mathcal{B}$. Write $R = F - F_\varepsilon$. Then $\|R\| < \eta$, and proposition 7.1 excludes exact labels on $\mathcal{E}$. The definition of $\mathcal{B}$ excludes exact labels everywhere else. Thus $\mathcal{R}_F(A) \neq 0$ for every $A \in \mathcal{O}_d$. By lemma 2.2, this is precisely the failure of (1.3). The polynomial $F \in \mathcal{P}_d$ is odd and has degree at most D , and its sphere restriction is smooth. This proves both theorems. $\square$

Remark 7.2. The argument is an existence proof in an explicitly specified finite-dimensional polynomial space. It does not exhibit coefficients for R , a numerical ε_0 , or an optimal degree bound. It also does not prove that $Q + \varepsilon H$ alone is a counterexample for every $d \geq 4$. The purpose of R is to avoid the nonexceptional placements without classifying all their cubic solutions.

8. THE CONSTANT-WIDTH AND UNIVERSAL-COVER CONSEQUENCE

The passage from odd functions to small perturbations of the support function of a ball is standard; see [K99, Section I, Proposition 1] and [Ch26, Lemma 8]. We include the needed implication with width normalized to one.

Lemma 8.1 (Realizing a small odd perturbation). *Let $f : S^{d-1} \rightarrow \mathbb{R}$ be smooth and odd. For all sufficiently small $\tau > 0$, there is a compact convex body K of constant width one whose support function on the sphere is*

$$h_K(w) = \frac{1}{2} + \tau f(w). \quad (8.1)$$

Proof. For $w \neq 0$ define

$$g(w) = \|w\| f\left(\frac{w}{\|w\|}\right), \quad \hat{h}(w) = \frac{1}{2}\|w\| + \tau g(w),$$

and set $\hat{h}(0) = 0$. Both functions are homogeneous of degree one for positive scalars. Differentiating Euler's identity gives $\nabla^2 g(w)w = 0$ and $\nabla^2 \hat{h}(w)w = 0$. At $\|w\| = 1$,

$$\nabla^2 \hat{h}(w) = \frac{1}{2}(I - ww^T) + \tau \nabla^2 g(w).$$

The Hessian of g is uniformly bounded on the unit sphere. Thus, for sufficiently small $\tau > 0$, the Hessian of $\hat{h}$ is positive definite on $w^\perp$ and positive semidefinite on $\mathbb{R}^d$ at each nonzero w . Here radial and mixed terms vanish, and homogeneity extends the assertion from the unit sphere to all $w \neq 0$.

The restriction of $\hat{h}$ to a line not passing through the origin is convex by the nonnegative second derivative. On a line through the origin, it is piecewise linear, and the right slope minus the left slope, using a unit direction w , is

$$\hat{h}(w) + \hat{h}(-w) = 1.$$

Hence it is convex on those lines as well. It follows that $\hat{h}$ is convex on all of $\mathbb{R}^d$.

Choose τ still smaller if needed so that $\tau\|f\|_\infty < 1/2$. Define

$$K = \{z \in \mathbb{R}^d : \langle z, w \rangle \leq \hat{h}(w) \text{ for every } w \in S^{d-1}\}.$$

This is closed and convex, contains a ball of positive radius, and is bounded by $\max_{S^{d-1}} \hat{h}$. To verify that its support function is exactly $\hat{h}$, fix $w \neq 0$. Convexity and differentiability at w imply

$$\hat{h}(y) \geq \hat{h}(w) + \langle \nabla \hat{h}(w), y - w \rangle = \langle \nabla \hat{h}(w), y \rangle,$$

where the last equality is Euler's identity. Thus $\nabla \hat{h}(w) \in K$, and $\langle \nabla \hat{h}(w), w \rangle = \hat{h}(w)$. The defining inequality of K gives the reverse support bound. Therefore $h_K = \hat{h}$ and (8.1) holds. Finally, oddness gives $h_K(w) + h_K(-w) = 1$ for every unit direction, which is constant width one. $\square$

Proof of Corollary 1.3. Take $f = F|_{S^{d-1}}$ from theorem 1.1, and form K as in lemma 8.1. Suppose $K \subset AC_d + b$. For each $\alpha = \alpha_{ij}$, the two defining half-spaces give

$$h_K(A\alpha) \leq \frac{1}{2} + \langle b, A\alpha \rangle, \quad h_K(-A\alpha) \leq \frac{1}{2} - \langle b, A\alpha \rangle.$$

The two left sides sum to one, as do the right sides. Both inequalities must therefore be equalities, so

$$\tau f(A\alpha) = \langle b, A\alpha \rangle \quad \text{for every positive root } \alpha.$$

Division by $\tau > 0$ contradicts theorem 1.1.

For completeness, a compact convex body of constant width one has diameter one. For distinct $x, y \in K$, use $w = (x - y)/\|x - y\|$ to get $\|x - y\| \leq h_K(w) + h_K(-w) = 1$. Conversely, support points in opposite directions have scalar separation one, so their distance is at least one. Thus this K is itself a set of diameter one not covered by any congruent copy of C_d . $\square$

APPENDIX A. VERIFICATION NOTES AND SUPPLEMENTARY ALGEBRA CHECKS

The central dependencies of the proof can be checked independently. Lemma 4.4 is a statement about real symmetric matrices; it uses neither the cubic nor the exceptional normal form. Lemma 4.5 applies it to the quadratic evaluation isomorphism. Proposition 4.6 then uses only the interpolation lemma and semialgebraic fiber dimensions. On the other side, proposition 6.1 is a consequence of the four-column orthonormality relations and the symmetric gradient matrix. The explicit sum in lemma 6.2 finishes the exceptional analysis. Compactness combines the two parts.

For checking lemma 4.4, the strict condition $\|c\| < 1$ and the separation of the eigenvalue clusters are essential: they ensure that any multiple eigenvalue of B is an unchanged double eigenvalue of S , where the residue in (4.5) vanishes. In the incidence argument, the potential c is *not* restricted to that unit ball. The bounded auxiliary parameter is used only to obtain (4.9).

For checking the normal form, the coefficients $\gamma = U - a$ in Case 1 and $\gamma = e - a$ in Case 2 follow from inner products, not just from subtracting the two special rows. The transformations in Case 2 include simultaneous conjugation, which reverses both H and the annihilating matrix and therefore preserves their pairing. The factor $(\sqrt{2})^{-5} = 1/(4\sqrt{2})$ in (6.16) is retained throughout.

The companion script `verify_algebra.py` checks, using exact symbolic arithmetic, the infinitesimal circle invariance of Q, H , the cubic expansion, the normalized quintic edge formula, the three nonzero edge-group sums, and their final simplification. It also checks several explicitly specified four-column frames satisfying the normal-form hypotheses. These are regression checks for formulas: neither finitely many test frames nor the script certifies exhaustiveness of the normal form or any dimension argument. No computational output is used as a premise of the proof. The supplied `README.md` gives build and audit commands.

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